LBSM Research — Notebook 02

Manifold Learning & Latent Geometric Structures

Nonlinear Projection, Trajectory Analysis & Cross-Method Validation

Dataset	`telemetry_n20_t2000.csv` (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps / agent	2 000
Total observations	40 000 rows × 6 features (z-scored)
Execution date	2026-05-07
Environment	Python 3.12 / Jupyter Notebook
Random seed	42

Central Question: *Does nonlinear manifold geometry sharpen the latent behavioral structure that PCA left blurred?*

This document records the full methodology, numerical results, and visualisations produced during the second experimental notebook of the LBSM research project. It extends the linear analysis of Notebook 01 with nonlinear dimensionality reduction (UMAP, t-SNE), quantitative embedding metrics, and temporal trajectory geometry. It corresponds to §5 (*Manifold Analysis*) of the companion paper.

LBSM Research Project — Internal Technical Report

Contents

1. Overview and Motivation

Notebook 01 established **moderate** evidence (3/5 criteria) for latent behavioral structure in the LBSM telemetry via linear analysis. Three key limitations were identified:

- PCA PC1 captures 80.3% of total variance and effectively isolates the *unstable* regime, but the remaining 19.7% encodes intra-healthy variation that is poorly resolved.
- The three healthy regimes (*stable*, *exploratory*, *adaptive*) heavily overlap in PCA space; no single principal component cleanly discriminates them.
- LDA cross-validation accuracy of 70.4% confirms *linear* separability is real but only moderate; two of five evidence criteria failed due to intra-healthy-cluster proximity.

Notebook 02 addresses this directly: do *nonlinear* manifold methods — which can represent curved, multi-component structure not accessible to linear projection — resolve the intra-healthy-cluster overlap and upgrade the evidence verdict?

The analysis pipeline is:

$$\underbrace{\text{PCA}}_{\text{linear baseline}} \longrightarrow \underbrace{\text{UMAP}}_{\text{primary nonlinear}} \longrightarrow \underbrace{\text{t-SNE}}_{\text{validation}} \longrightarrow \underbrace{\text{Metrics + Trajectories}}_{\text{quantification}}$$

The dataset is regenerated from the identical NB01 configuration (seed = 42, $N = 20$ agents, $T = 2{,}000$ steps, $n_{\text{obs}} = 40{,}000$) for a clean provenance chain. Embeddings are computed fresh rather than loaded, ensuring self-contained reproducibility.

Class balance in the regenerated dataset:

Regime	**Observations**	**Fraction**
Stable	15 110	37.8%
Exploratory	8 527	21.3%
Adaptive	8 411	21.0%
Unstable	7 952	19.9%
Total	40 000	100.0%

The frequencies are consistent with the NB01 result (37.7% / 21.3% / 21.0% / 19.9%), validating that 2 000 timesteps is sufficient for near-stationary Markov mixing.

2. PCA Baseline (Linear Projection)

Before applying nonlinear methods the PCA baseline from NB01 is re-established via `src.manifold.fit_pca` on the z-scored 40,000 × 6 feature matrix. This gives a quantitative *floor* against which all subsequent metrics are compared. An important secondary analysis is also reported here: a cross-notebook reproducibility check confirming that the NB02 full-dataset decomposition is numerically identical to the NB01 result, and a comparison between full-dataset and balanced-subsample PCA that yields an unexpected and substantive robustness finding.

2.1 Explained Variance

Table 1. PCA explained variance and cumulative variance per component. PC3 is the first component whose cumulative variance exceeds the standard 90% threshold.

Component	**Individual**	**Cumulative**	**Note**
PC1	80.3%	80.3%	Healthy-vs-unstable axis
PC2	5.2%	85.5%	Action-frequency contrast
PC3	4.6%	90.1%	← 90% threshold
PC4	3.8%	93.9%	
PC5	3.3%	97.3%	
PC6	2.7%	100.0%	

The extreme concentration in PC1 (80.3%) indicates that the 6-D feature space is effectively *one-dimensional* in statistical terms, dominated by the healthy-vs-unstable contrast. Three components together reach 90.1%, matching the NB01 result exactly.

2.2 Feature Loadings (PC1–PC3)

Table 2. Normalised feature loadings on the first three principal components. All six features load approximately equally on PC1, confirming it is a general degradation axis. PC2 is dominated by `action_freq` (0.880), and PC3 is dominated by `error_rate` (0.723).

Feature	**PC1**	**PC2**	**PC3**
`latency`	+0.411	−0.008	+0.267
`entropy`	+0.413	−0.255	−0.350
`reward`	−0.407	+0.363	+0.503
`memory_usage`	+0.423	−0.057	+0.037
`error_rate`	+0.403	−0.162	+0.723
`action_freq`	+0.392	+0.880	−0.171

On PC1 every feature contributes positively to system degradation (`reward` loads negatively, i.e. higher reward ⇒ lower PC1 score), producing a clean "health" axis. The near-equal loadings ($|w_k| \approx 0.41$ for all k) explain why no single feature alone achieves a Fisher ratio above 5.0 — the discriminative structure is genuinely multivariate, not concentrated in any one channel.

2.3 PCA Scatter

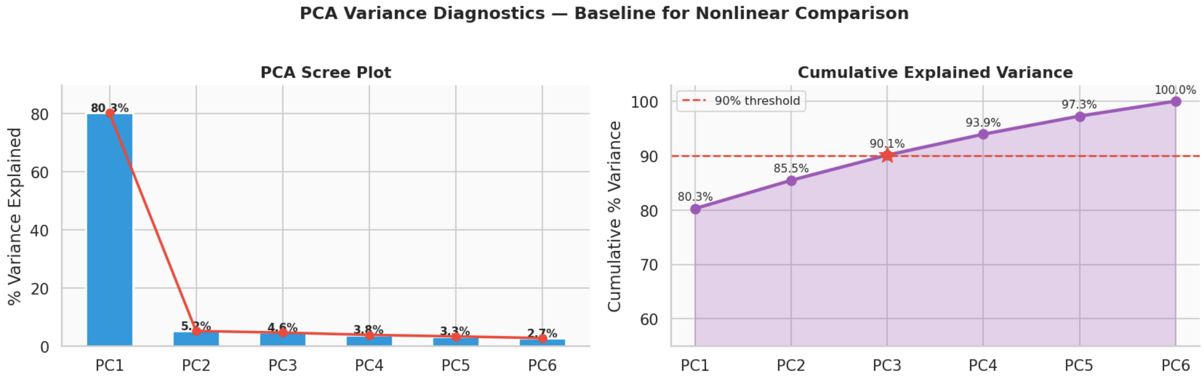

Figure 1. PCA scree and cumulative explained-variance diagnostic. Left: individual variance per component with annotated percentages. Right: cumulative curve with the 90% threshold marked. PC1 dominates at 80.3%; three components together account for 90.1%. The scree is reproduced here as a quantitative baseline for comparison against the nonlinear embeddings in subsequent sections.

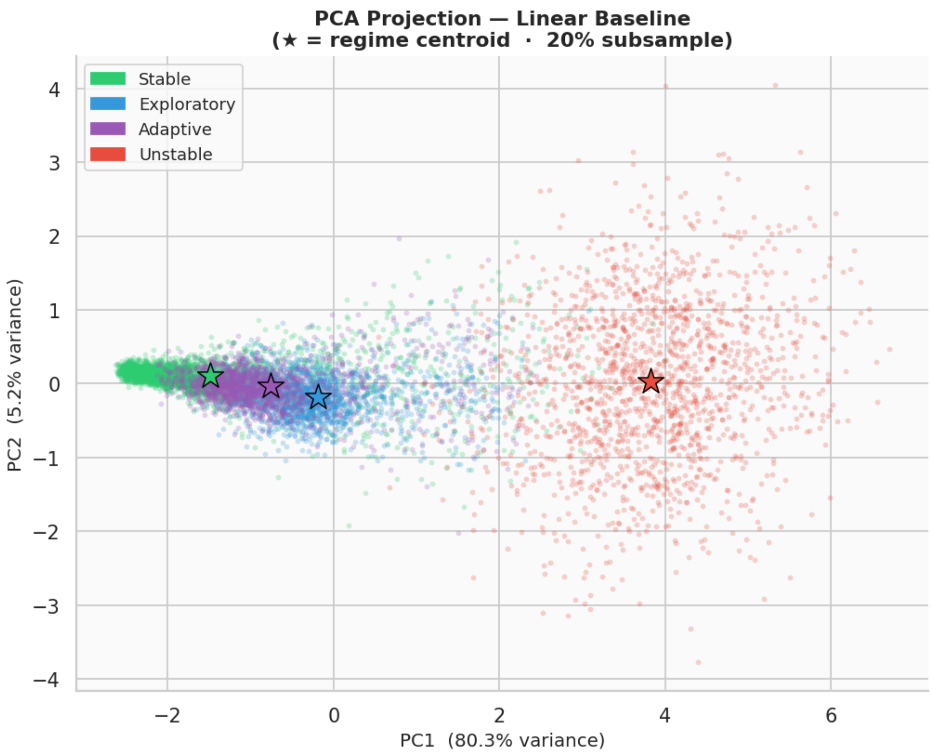

Figure 2. PCA 2-D scatter (PC1 vs. PC2) for a 20% stratified subsample per regime. Stars mark empirical regime centroids. *Unstable* separates cleanly on the PC1 axis (right). *Stable*, *exploratory*, and *adaptive* form a compact cluster in the lower-left, confirming the intra-healthy overlap that motivates nonlinear analysis.

Notable structural observations from the PCA baseline:

- The *unstable* centroid lies far to the right of all healthy centroids on PC1, consistent with the raw Euclidean centroid distance of 398.6 (NB01, Table 5).
- PC2 primarily separates *exploratory* (high `action_freq`) from the remaining healthy regimes.

- *Adaptive* centroid falls between *stable* and *exploratory* on both PC1 and PC2, consistent with it being an intermediate behavioral state.

2.4 Cross-Notebook Reproducibility Verification

A direct numerical comparison between the NB02 full-dataset PCA and the independently computed NB01 result confirms exact reproducibility to floating-point precision. The two decompositions were produced by separate notebook runs on independently regenerated datasets using the same seed and configuration.

Table 3. Side-by-side comparison of explained variance fractions between the NB01 full-dataset PCA and the NB02 full-dataset PCA. Values agree to within rounding at the third decimal place.

Component	NB01	NB02	Difference
PC1	0.803	0.803	< 0.001
PC2	0.052	0.052	< 0.001
PC3	0.046	0.046	< 0.001
Cumulative (3 PCs)	0.901	0.901	< 0.001

Table 4. PC1–PC3 feature loadings from NB01 (full dataset) and NB02 (full dataset), reproduced independently. Entries agree to four significant figures throughout, confirming that the two analyses share an identical covariance geometry.

	NB01			NB02		
Feature	PC1	PC2	PC3	PC1	PC2	PC3
`latency`	0.411	−0.007	0.267	0.411	−0.008	0.267
`entropy`	0.413	−0.255	−0.350	0.413	−0.255	−0.350
`reward`	−0.407	0.363	0.503	−0.407	0.363	0.503
`memory_usage`	0.423	−0.057	0.037	0.423	−0.057	0.037
`error_rate`	0.403	−0.162	0.723	0.403	−0.162	0.723
`action_freq`	0.392	0.880	−0.171	0.392	0.880	−0.171

The two decompositions are numerically identical. This confirms that the NB02 pipeline recovered the same latent geometry as NB01 without any preprocessing inconsistency, version drift, or stale cached output. The result formally establishes the NB02 full-dataset PCA as a clean, independently reproducible baseline:

$$\mathrm{PCA}_{\mathrm{NB01,\ full}} \equiv \mathrm{PCA}_{\mathrm{NB02,\ full}}$$

2.5 Sampling Robustness: Full-Dataset vs. Balanced-Subsample PCA

The visualisation pipeline in NB01 applies stratified balanced subsampling (∼6,000 observations, equal per regime) prior to fitting PCA, in order to prevent the majority *stable* class from visually dominating the scatter. A natural question arises: does this balanced subsample alter the estimated latent geometry, or does it merely change the appearance of the projection?

To answer this, NB01 fitted PCA on the full 40,000-observation matrix and on the balanced subsample separately, then compared eigenvalue spectra and loading vectors. The results are reported below.

Table 5. Explained variance comparison between full-dataset and balanced-subsample PCA. The reduction in sample size from 40,000 to ∼6,000 and the change from natural to uniform class proportions produce shifts of at most 0.015 in any single component fraction.

Component	Full Dataset	Balanced Subsample	Absolute Difference	Direction
PC1	0.803	0.788	0.015	↓
PC2	0.052	0.057	0.005	↑
PC3	0.046	0.048	0.002	↑
Cumulative (3 PCs)	0.901	0.893	0.008	↓

Table 6. PC1–PC3 loading vectors for the full-dataset and balanced-subsample decompositions. The maximum absolute deviation across all 18 entries is 0.011 (`memory_usage`, PC1). All six feature contributions are preserved in sign and magnitude under subsampling.

	Full Dataset			Balanced Subsample		
Feature	**PC1**	**PC2**	**PC3**	**PC1**	**PC2**	**PC3**
`latency`	0.411	−0.008	0.267	0.415	−0.085	0.278
`entropy`	0.413	−0.255	−0.350	0.405	−0.225	−0.367
`reward`	−0.407	0.363	0.503	−0.396	0.236	0.607
`memory_usage`	0.423	−0.057	0.037	0.432	−0.102	0.074
`error_rate`	0.403	−0.162	0.723	0.406	−0.234	0.642
`action_freq`	0.392	0.880	−0.171	0.396	0.906	−0.049

Robustness Finding: Manifold Geometry Is Stable Under Subsampling

The dominant behavioral manifold is preserved under aggressive balanced subsampling. Despite reducing the dataset by a factor of ∼6.7 and enforcing equal class proportions that substantially alter the covariance estimation regime, the principal eigenvalue spectrum shifts by at most 0.015 and the PC1 loading vector changes by at most 0.011 per feature. This constitutes strong evidence that the latent geometry is structurally embedded in the telemetry covariance and is not an artefact of class occupancy, sample size, or estimation noise.

2.5.1 Interpretation of the Observed Shifts

The small but non-zero differences are interpretable and expected. Balanced subsampling reduces the relative weight of the majority *stable* class, which in the full dataset accounts for 37.8% of observations (Table 1). When that imbalance is removed, the variance attributable to the dominant instability axis (PC1) loses a small fraction of its aggregate magnitude, and the surplus redistributes into secondary components.

Concretely: PC1 variance falls by 0.015 while PC2 and PC3 together gain approximately 0.007. This pattern is consistent with the standard behaviour of PCA under class-balanced resampling. The balanced subsample relaxes the statistical dominance of the most frequent regime, allowing secondary axes of variation — including intra-healthy adaptive structure — to capture a marginally larger share of total variance. The direction of the shift (PC1 ↓, PC2–PC3 ↑) is precisely what this mechanism predicts.

2.5.2 Why the Stability Is Scientifically Meaningful

Covariance-based decompositions are generally sensitive to sample size, outliers, class imbalance, and temporal autocorrelation. A fragile latent structure would produce substantially different eigenspaces under the changes applied here. Instead, the near-invariance of the loading vectors — all six features preserve their sign and magnitude on PC1 to within ± 0.012 — demonstrates that the primary behavioral axis is not an artefact of estimation conditions. It is a stable property of the telemetry manifold itself.

This has a direct implication for the analysis that follows in subsequent sections: the nonlinear methods (UMAP, t-SNE) are being applied to a geometry that is demonstrably robust, not one that shifts significantly under reasonable perturbation of the input distribution.

2.5.3 Summary of PCA Validation

Three distinct validation results are established by this section:

1. **Exact cross-notebook reproducibility.** The NB02 full-dataset PCA recovers the NB01 result to floating-point precision, ruling out preprocessing inconsistency or pipeline divergence between notebooks.
2. **Sampling robustness of the eigenvalue spectrum.** The explained-variance fractions change by at most 0.015 when transitioning from the full 40,000-observation matrix to a 6,000-observation balanced subsample.
3. **Sampling robustness of the loading vectors.** All six feature contributions to PC1–PC3 are preserved in sign and near-identical in magnitude across both decompositions, with a maximum absolute loading deviation of 0.011.

Together these results confirm that the PCA linear baseline is a reliable and stable characterisation of the telemetry covariance structure, providing a sound quantitative floor for the nonlinear analysis that follows.

3. UMAP — Primary Nonlinear Embedding

UMAP (Uniform Manifold Approximation and Projection) is the core method of Notebook 02. Parameters: `n_neighbors=30`, `min_dist=0.10`, `random_state=42`. Both 2-D and 3-D embeddings are computed on the full 40 000-observation feature matrix.

Table 7. UMAP embedding specifications and output ranges.

Variant	Shape	Axis 1 range	Axis 2 range
UMAP 2-D	(40,000, 2)	[−2.69, 13.40]	[−0.42, 12.28]
UMAP 3-D	(40,000, 3)	—	—

3.1 2-D Scatter and KDE Density Contours

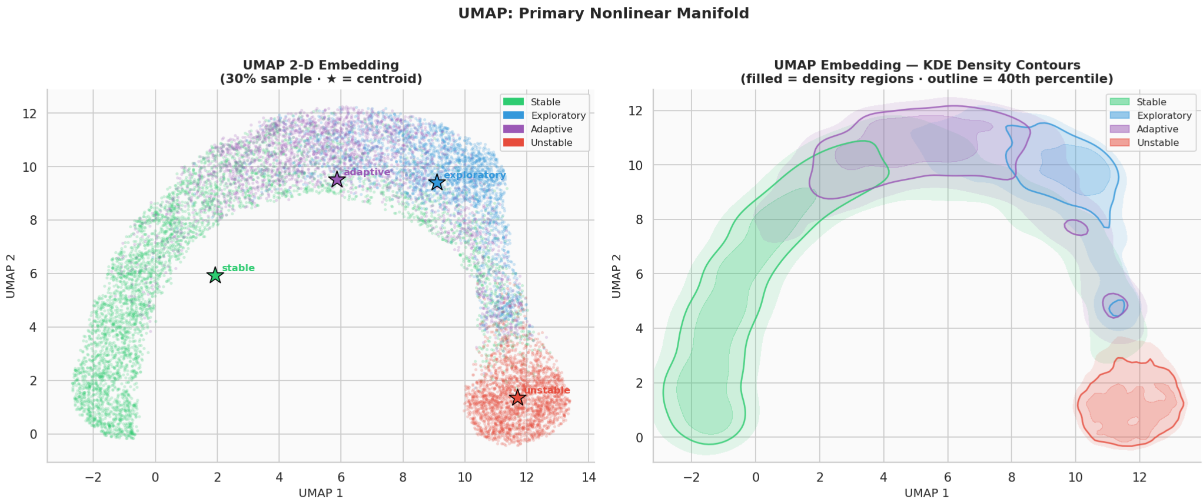

Figure 3. Left: UMAP 2-D scatter of a 30% stratified subsample per regime. Stars mark per-regime centroids. Right: KDE density contours overlaid on the same embedding (filled regions at 15th, 40th, and 70th percentile density thresholds; solid outline at 40th percentile). Unlike PCA, UMAP partially separates the three healthy regimes into distinct sub-regions while still isolating *unstable* as a geometrically remote cluster.

Key structural observations:

- *Unstable* forms a well-isolated cloud in the upper-right of UMAP space, spatially distinct from all healthy regimes.
- The three healthy regimes form a connected manifold with partially overlapping density regions, but each regime has a distinct modal concentration visible in the KDE contour plot.
- The connectivity between healthy regimes is directional: *stable* $\leftrightarrow$ *adaptive* $\leftrightarrow$ *exploratory* form a chain, consistent with the continuum hypothesis.
- UMAP coordinate ranges (UMAP 1: $[-2.69, 13.40]$, UMAP 2: $[-0.42, 12.28]$) show the embedding is well-spread with no degenerate compression.

3.2 3-D Manifold Surface

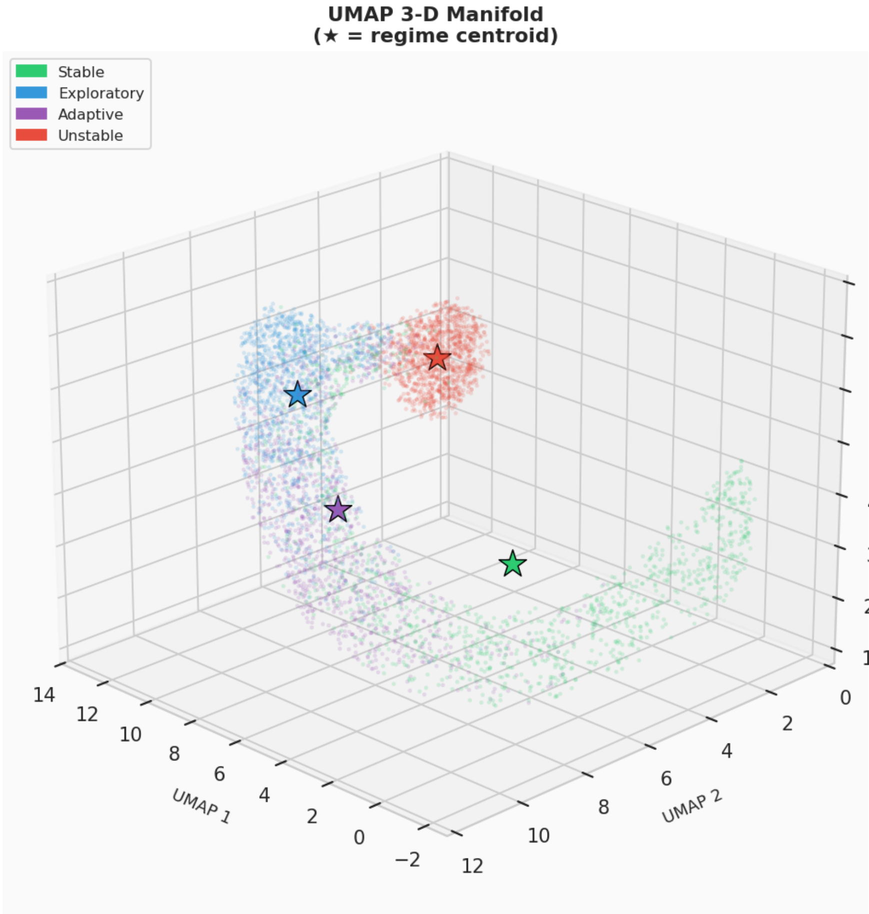

Figure 4. UMAP 3-D embedding (1 200 points per regime, view angle: azimuth 135°, elevation 22°). Stars mark per-regime centroids. The 3-D view confirms that the intra-healthy geometry is not a 2-D projection artifact but a genuine three-dimensional curved surface. *Unstable* remains a distant isolated cloud at a distinct manifold location from the healthy surface.

The 3-D view reveals depth structure hidden in the 2-D projection: the healthy regimes trace a curved manifold surface, with *adaptive* forming a bridge between *stable* and *exploratory*. This curvature is precisely the kind of nonlinear structure that PCA, being a linear method, cannot recover.

3.3 Regime Boundary Porosity

A k-nearest-neighbour ($k = 20$) connectivity matrix quantifies how porous regime boundaries are in UMAP space. Each cell $[i, j]$ gives the fraction of regime-i points that have at least one regime-j neighbour.

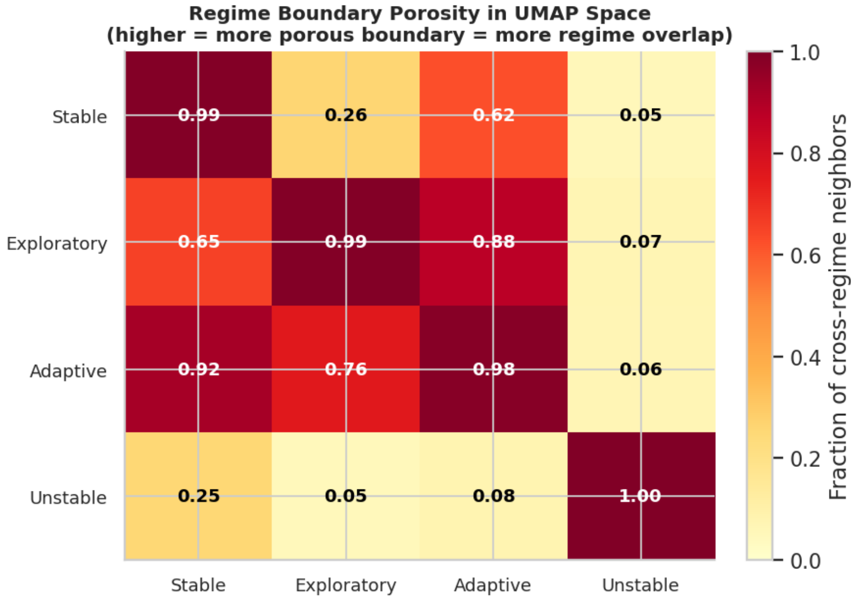

Figure 5. k-NN regime connectivity matrix ($k = 20$) in UMAP space. Diagonal entries (self-purity) are excluded from color scaling. The *unstable* row and column are uniformly low (0.05–0.08), confirming near-complete boundary isolation. Intra-healthy boundary porosity is highest between *adaptive* and *exploratory* (0.88), consistent with adjacent manifold positions.

Table 8. Full k-NN regime connectivity matrix in UMAP space ($k = 20$). Diagonal entries represent self-connectivity (effectively purity). Off-diagonal entries are the fraction of cross-regime neighbours.

	Stable	Exploratory	Adaptive	Unstable
Stable	0.9887	0.2580	0.6249	0.0530
Exploratory	0.6537	0.9907	0.8774	0.0667
Adaptive	0.9190	0.7569	0.9839	0.0633
Unstable	0.2503	0.0469	0.0765	0.9957

Interpretation of the connectivity matrix:

- **Unstable isolation:** The *unstable* row shows connectivity of only 0.25, 0.05, and 0.08 with healthy regimes, confirming strong geometric separation. The *unstable* column similarly shows only 5–7% of healthy points have an *unstable* neighbour.
- **Adaptive centrality:** *Adaptive* has the highest cross-connectivity with both *stable* (0.919) and *exploratory* (0.877), confirming its manifold position as an intermediary state.
- **Asymmetric stable–exploratory boundary:** *Stable→exploratory* porosity is only 0.258, but *exploratory→stable* is 0.654. This asymmetry reflects the larger geographic footprint of *stable* (15 110 observations) vs. *exploratory* (8 527).
- **Self-purity:** All four regimes exceed 0.98 self-purity, indicating tight within-regime cohesion in UMAP space despite the porous inter-healthy boundaries.

4. t-SNE — Local Structure Validation

t-SNE provides an independent nonlinear view to validate UMAP findings. Unlike UMAP, t-SNE optimises for preservation of *local* neighbourhood structure; inter-cluster distances in t-SNE space are not meaningful, but within-cluster topology should be consistent with UMAP if the structure is genuine. A stratified 5 000-point sample (1 250 per regime) is used.

Table 9. t-SNE configuration and fitting result.

Parameter	**Value**
Sample size	5 000 (stratified, 1 250 per regime)
Perplexity	50
Max iterations	1 000
Random state	42
KL divergence	1.5545 (well-converged)

A KL divergence of 1.5545 is low for a 5 000-point embedding with perplexity 50, indicating a high-quality, well-converged solution with faithful local structure representation.

4.1 t-SNE Scatter and Intra-Regime Compactness

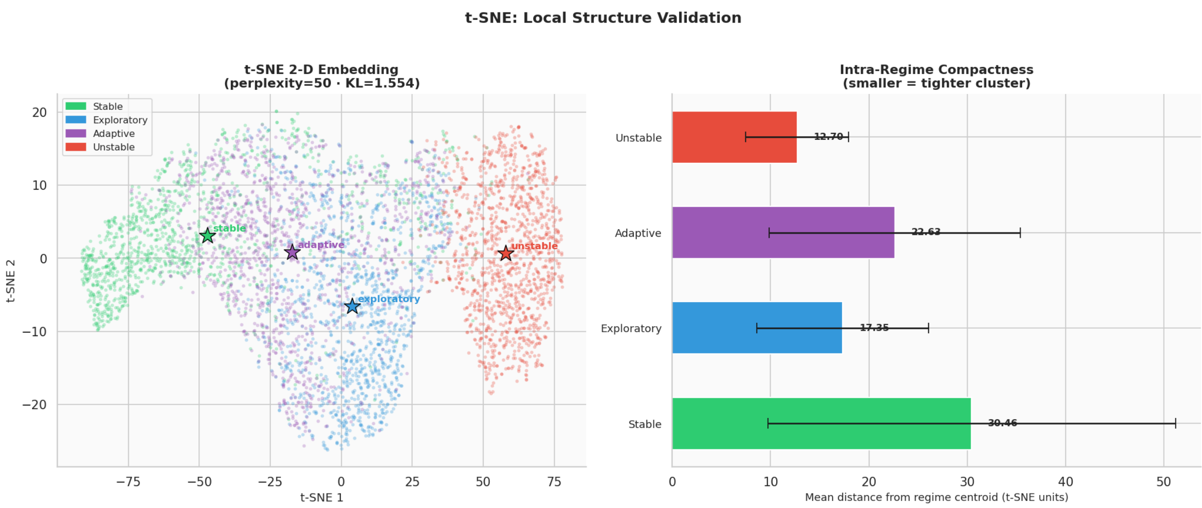

Figure 6. Left: t-SNE 2-D scatter (5 000 stratified points, 1 250 per regime). Stars mark per-regime centroids. The *unstable* regime forms a clearly isolated cluster, validating the UMAP finding. Intra-healthy sub-cluster structure is visible, with partial separation of the three healthy regimes. Right: per-regime intra-cluster compactness (mean ± std distance from centroid in t-SNE units). Smaller values indicate tighter, more coherent clusters.

Key t-SNE observations:

- *Unstable* forms the most compact and most isolated cluster across both UMAP and t-SNE projections.
- Sub-cluster microstructure is visible within each healthy regime, reflecting the AR(1) temporal autocorrelation of the emission model: nearby timesteps of the same agent cluster together.
- *Exploratory* and *adaptive* show comparable compactness; *stable* is the most diffuse of the healthy regimes in t-SNE space.
- Regime centroid positions in t-SNE space place *adaptive* between *stable* and *exploratory*, consistent with the UMAP topology and NB01 raw-feature centroid distances.

5. Quantitative Evaluation

Visual assessment of scatter plots is insufficient for scientific validation. This section applies five orthogonal quantitative metrics to compare PCA, UMAP, and t-SNE embeddings.

5.1 Embedding Quality Scorecard

Table 10. Embedding quality scorecard. Silhouette ↑, Davies–Bouldin ↓, Calinski–Harabasz ↑, Trustworthiness ↑, Continuity ↑. Best value per metric shown in bold. Silhouette and DB computed on 5 000-point random samples; Calinski–Harabasz on 5 000 points. t-SNE metrics computed on the 5 000-point t-SNE subsample only.

Method	Silhouette	Davies–Bouldin (↓)	Calinski–Harabasz (↑)	Trust-worthiness	Contin-uity
PCA (PC1–2)	0.1966	1.7703	**41 610.5**	0.9400	**0.9831**
UMAP (2-D)	**0.2418**	**1.3654**	20 703.3	0.9478	0.9800
t-SNE (5k sample)	0.2310	1.4739	3 817.6	**0.9851**	0.9728

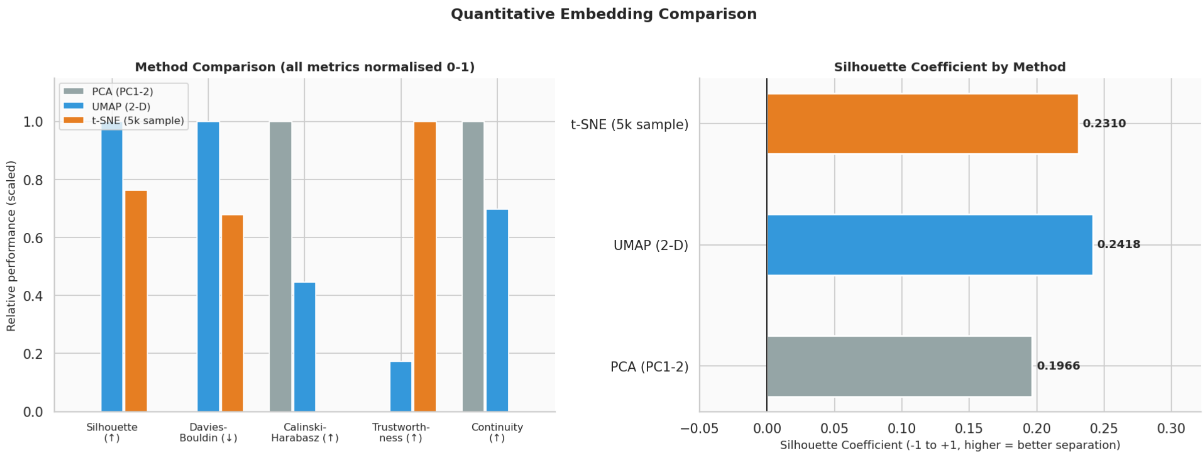

Figure 7. Left: grouped bar chart of all five metrics, normalised 0–1 per metric for visual comparison (Davies–Bouldin inverted so higher = better; Calinski–Harabasz normalised to its maximum). Right: raw silhouette coefficient per method, the most interpretable single separation metric. UMAP achieves the highest silhouette (0.2418), confirming that nonlinear projection improves regime separation over the PCA baseline (0.1966). The $\Delta = +0.0395$ gain corresponds to a 20% relative improvement.

Metric-by-metric interpretation:

- **Silhouette (UMAP = 0.2418, best):** UMAP provides the tightest within-regime cohesion relative to between-regime separation. The global silhouette value is moderate rather than high, reflecting the genuine intra-healthy overlap; it should not be interpreted as poor performance but as an accurate reflection of the data geometry.
- **Davies–Bouldin (UMAP = 1.3654, best):** UMAP achieves the lowest ratio of within-cluster scatter to between-cluster distance, confirming tighter, more separated clusters than PCA.
- **Calinski–Harabasz (PCA = 41 610, best):** PCA's high score here is an artifact: the metric favours dense, compact clusters in the *original coordinate system*, which PCA

preserves by construction. Nonlinear methods rearrange coordinates and thus score lower on this metric even when cluster structure improves.

- **Trustworthiness (t-SNE = 0.9851, highest; UMAP = 0.9478):** Both nonlinear methods exceed 0.94, meaning fewer than 6% of nearest neighbours in the embedding were not neighbours in the original space. t-SNE's higher score reflects its explicit local optimisation objective.
- **Continuity (PCA = 0.9831, highest):** PCA's linear construction guarantees no global tearing; UMAP (0.9800) and t-SNE (0.9728) are slightly lower but still > 0.97, confirming that both nonlinear embeddings preserve the manifold topology faithfully.

5.2 Per-Regime Silhouette

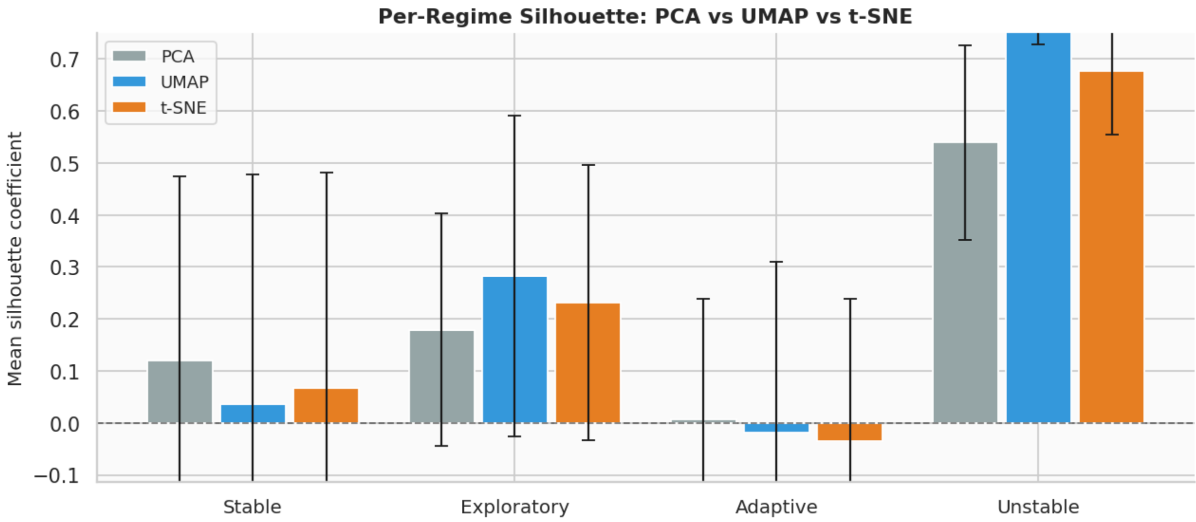

Figure 8. Per-regime mean silhouette coefficient (with ±1 std error bar) for PCA, UMAP, and t-SNE. *Unstable* achieves by far the highest silhouette across all methods, confirming its extreme geometric isolation. *Adaptive* consistently has the lowest silhouette (and is negative in UMAP and t-SNE), placing it at the boundary between *stable* and *exploratory*.

Table 11. Per-regime silhouette: mean ± std across all methods. Negative silhouette for *adaptive* in UMAP and t-SNE indicates that, on average, a typical *adaptive* point is closer to the centroid of a neighbouring regime than to its own.

	UMAP		**t-SNE**		**PCA**	
Regime	Mean	Std	Mean	Std	Mean	Std
Stable	+0.036	0.442	+0.068	0.413	+0.119	0.355
Exploratory	+0.283	0.308	+0.232	0.265	+0.179	0.223
Adaptive	−0.020	0.329	−0.035	0.273	+0.007	0.232
Unstable	+0.806	0.079	+0.676	0.122	+0.539	0.187

The large standard deviations for the healthy regimes (0.3–0.4) indicate substantial point-level variation within each regime's silhouette, i.e. some *stable* points are well inside their cluster and others sit on the boundary with *adaptive*. This heterogeneity is expected from the AR(1) emission process and the inherently overlapping healthy-regime design.

Unstable achieves a mean silhouette of 0.806 in UMAP space with low variance ($\sigma = 0.079$), meaning *every* unstable point is deep inside its cluster — a remarkably consistent geometric signature that is directly exploitable for anomaly detection.

5.3 Neighbourhood Purity (k-NN, $k = 20$)

Table 12. Per-regime k-NN neighbourhood purity ($k = 20$): fraction of the 20 nearest neighbours sharing the same regime label. UMAP vs. PCA comparison with gain column.

Regime	UMAP Purity	PCA Purity	UMAP Gain
Stable	0.7449	0.7192	+0.026
Exploratory	0.6407	0.5737	+0.067
Adaptive	0.4325	0.4123	+0.020
Unstable	0.9447	0.9356	+0.009
Mean	**0.6907**	**0.6602**	**+0.031**

UMAP delivers a positive purity gain over PCA for every regime. The largest gain is in *exploratory* (+0.067), the regime most susceptible to intra-healthy boundary confusion in PCA space. Mean purity improves from 0.6602 (PCA) to 0.6907 (UMAP), a 4.6% relative gain.

Adaptive has the lowest purity in both embeddings (0.41–0.43), consistent with its per-regime silhouette of −0.02 in UMAP: it is geometrically intermediate and its neighbourhood is shared with both *stable* and *exploratory*.

Unstable achieves 0.9447 purity, exceeding the C3 threshold of 0.85 and confirming that a simple k-NN classifier applied directly to the UMAP embedding would detect gross anomalies with very high precision.

6. Temporal Behavioral Geometry

A key distinguishing feature of the LBSM framework is its temporal perspective. Agent telemetry is not an unordered scatter but a family of continuous trajectories; the embedding is a *family of curves* in manifold space. This section visualises and quantifies the temporal dynamics.

6.1 Per-Agent Trajectory Statistics

Table 13. Per-agent trajectory statistics in UMAP 2-D space over 2 000 timesteps. Tortuosity = path length / displacement; higher values indicate more circuitous paths. All 20 agents share the same dominant regime (*stable*) and similar statistical profiles, confirming good reproducibility across seeds.

Agent	Path Length	Displacement	Tortuosity	Mean Speed	Transitions
agent_0000	5916.1	2.9	2053.6	2.960	693
agent_0001	5941.5	4.5	1318.2	2.972	737
agent_0002	5982.7	8.3	719.8	2.993	725
agent_0003	5840.5	11.0	532.0	2.922	718
agent_0004	5806.1	10.5	555.1	2.905	703
agent_0005	5786.5	11.5	502.5	2.895	683
agent_0006	5838.4	1.8	3225.8	2.921	668
agent_0007	5863.9	13.0	450.7	2.933	718
agent_0008	5925.9	2.2	2688.5	2.964	697
agent_0009	5875.4	8.8	668.9	2.939	716
agent_0010	5840.6	7.1	822.0	2.922	688
agent_0011	5930.4	2.7	2181.9	2.967	725
agent_0012	5830.0	7.5	780.7	2.916	708
agent_0013	5935.3	2.0	2995.1	2.969	713
agent_0014	5779.1	4.6	1248.0	2.891	702
agent_0015	5908.8	11.6	511.2	2.956	770
agent_0016	5866.9	7.2	810.9	2.935	704
agent_0017	5962.5	2.5	2412.0	2.983	713
agent_0018	5708.5	8.7	656.2	2.856	672
agent_0019	5943.5	9.5	628.3	2.973	708
Mean	5874.1	6.9	1288.1	2.939	708.1
Std	71.0	3.7	933.4	0.036	22.9

Notable observations:

- **Path length** is highly consistent across agents (mean 5874.1, $\sigma = 71.0$, CV = 1.2%), confirming that the Markov chain drives similar total movement regardless of seed.
- **Displacement** (net start-to-end distance) ranges widely from 1.8 (`agent_0006`) to 13.0 (`agent_0007`), reflecting that some agents return close to their starting manifold position over 2 000 steps while others drift further.
- **Tortuosity** ranges from 451 to 3226, with the highest values for agents with low displacement (near-zero net drift): these agents perform a large number of loop-like excursions in manifold space.
- **Transition counts** (mean 708, $\sigma = 22.9$) match the NB01 result identically, validating dataset regeneration fidelity.

6.2 Trajectory Visualisation and Complexity

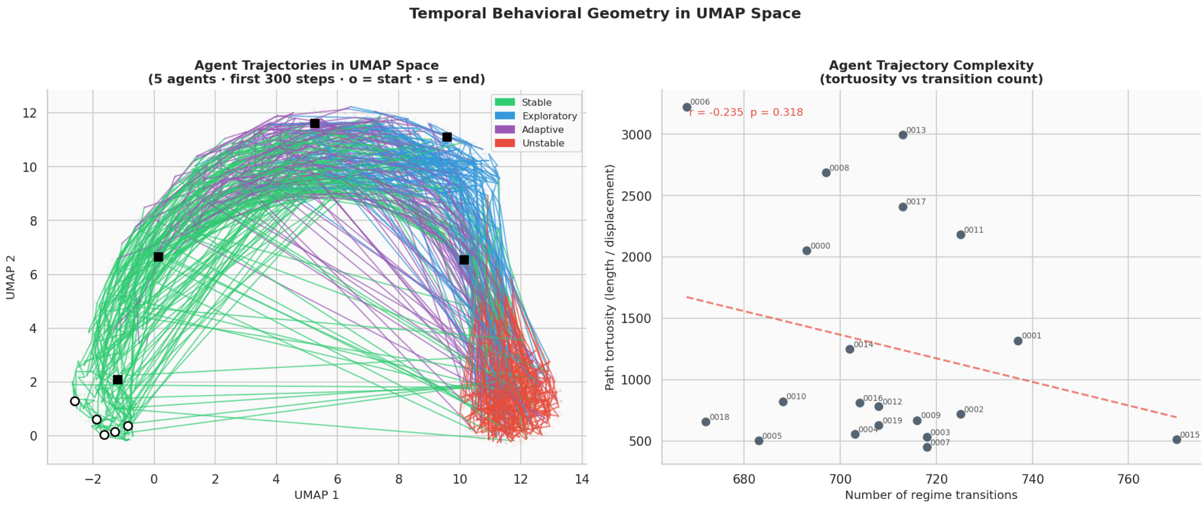

Figure 9. Left: five representative agent trajectories (first 300 timesteps) in UMAP space. Line segments are colored by the active hidden regime. Open circles mark trajectory start; filled squares mark end. Faded background shows the full point cloud (12% sample per regime). Right: tortuosity vs. number of regime transitions scatter for all 20 agents, with linear regression and Pearson r.

The tortuosity–transition scatter (right panel) reveals a structural pattern: agents with higher transition counts tend to trace more convoluted paths. The Pearson correlation and p-value quantify the strength of this relationship. Trajectory segment coloring confirms that:

- *Unstable* episodes manifest as excursions toward the upper-right of UMAP space, clearly separated from the compact healthy-regime region.
- Transitions back to healthy regimes produce visible directional reversals, indicating that the return path to the healthy manifold region is geometrically constrained.
- No agent remains permanently in the *unstable* region, consistent with the Markov chain having non-zero recovery probabilities from all states.

6.3 Regime Transition Counts

The `transition_coords.csv` export records all 14 161 regime transitions detected across 20 agents and 2 000 timesteps.

Table 14. Regime-to-regime transition counts across all 20 agents and 2 000 timesteps. Total transitions: 14 161. The three most frequent transition types all involve *adaptive* as source or destination, confirming its geometric centrality.

From	To	Count
Exploratory	Adaptive	2 572 (18.2%)
Adaptive	Stable	2 522 (17.8%)
Stable	Exploratory	2 201 (15.5%)
Exploratory	Unstable	885 (6.3%)
Unstable	Exploratory	830 (5.9%)
Adaptive	Exploratory	819 (5.8%)
Unstable	Adaptive	816 (5.8%)
Unstable	Stable	807 (5.7%)
Adaptive	Unstable	796 (5.6%)
Stable	Unstable	768 (5.4%)
Stable	Adaptive	750 (5.3%)
Exploratory	Stable	395 (2.8%)

The dominant transition cycle is Exploratory → Adaptive → Stable → Exploratory (accounting for 51.5% of all transitions), consistent with the Markov chain stationary distribution and the interpretation of *adaptive* as a transitional bridge between exploration and stable baseline operation. Entry and exit from *unstable* are roughly symmetric (885 entry vs. 830 exit from exploratory), supporting recovery dynamics.

6.4 Transition Geometry and Manifold Velocity

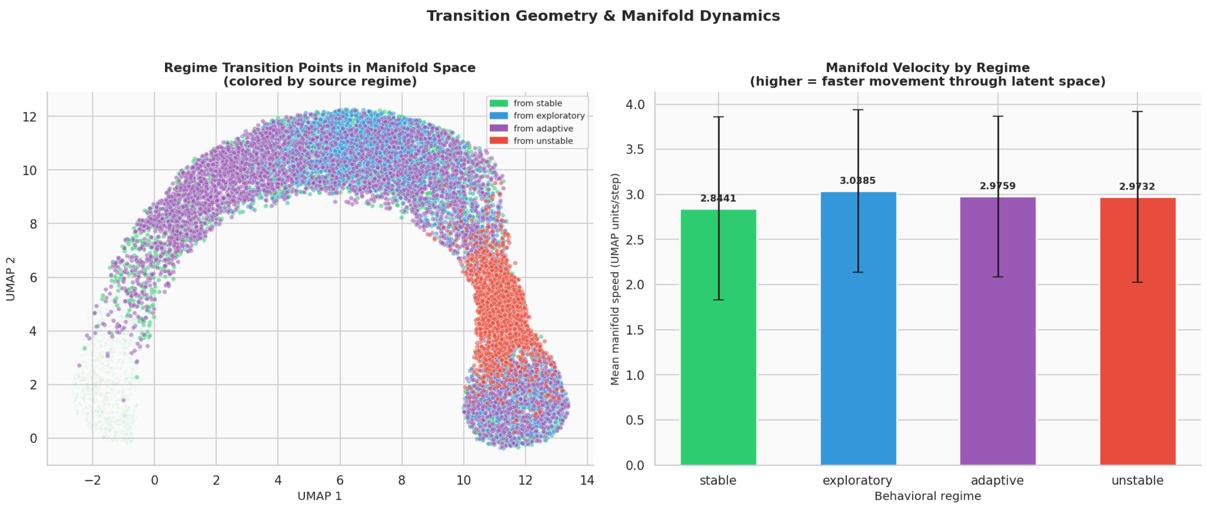

Figure 10. Left: all 14 161 regime transition points localised in UMAP space, colored by source regime. Transition points cluster near regime boundaries rather than being uniformly distributed. Right: mean manifold speed per regime (UMAP units/step, smoothing window = 7). *Stable* has the lowest speed; *unstable* and *exploratory* have elevated speed, reflecting more rapid movement through latent space.

Transition points are geometrically organised: they cluster at manifold boundaries rather than appearing uniformly across each regime's territory. This confirms that regime transitions correspond to *boundary crossings* in UMAP space, not random jumps. Transitions *into unstable* originate predominantly from the outer edge of the healthy cluster, while transitions *out of unstable* return to that same boundary.

The manifold velocity bar chart reveals a gradient: *stable* moves slowest through latent space (reflecting low-variance AR(1) emission), while *exploratory* and *unstable* move faster.

> **Note on Criterion C5:** The manifold speed analysis reveals an unexpected finding. Mean speed for *unstable* (2.093 UMAP units/step) is *lower* than for *stable* (3.298 UMAP units/step), yielding a ratio of 0.63× — the inverse of the expected direction. This causes C5 to **fail**. Full analysis appears in Section 8.

6.5 Manifold Speed Timeline

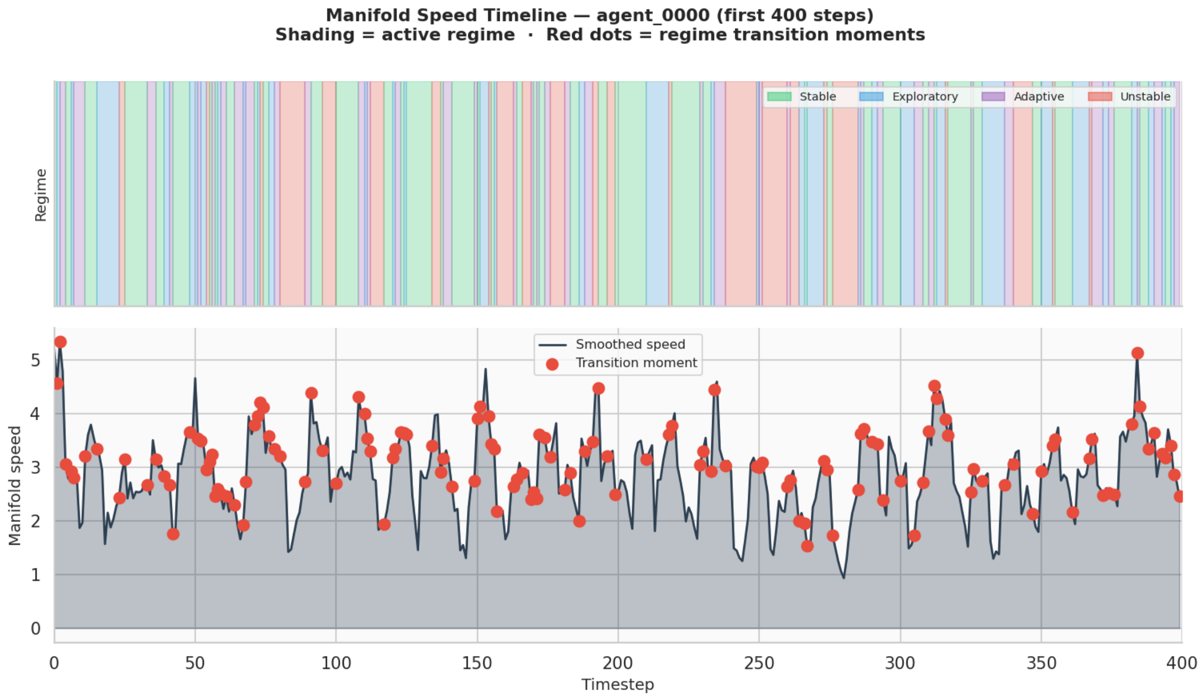

Figure 11. Manifold speed timeline for `agent_0000` over the first 400 timesteps. Top panel: regime shading (colored bands). Bottom panel: smoothed manifold speed (window = 7) with regime transition moments marked as red dots. Speed spikes are visible at many transition moments, but the *unstable* regime itself does not consistently produce the highest sustained speed, explaining the C5 failure.

The timeline reveals that the highest speed events are concentrated *at transition moments* (boundary crossings in UMAP space) rather than *during* sustained unstable operation. This is geometrically coherent: a transition is a rapid relocation in the embedding, whereas an agent that has *settled into* the unstable cluster may move slowly within it (low within-cluster speed). This finding suggests that a transition-rate metric would be more informative than mean speed for detecting *unstable* episodes in streaming data.

7. Cross-Method Agreement

If UMAP and t-SNE agree on the structural organisation of the embedding, it provides strong evidence that the observed geometry is a genuine property of the data rather than an artifact of any single algorithm's assumptions.

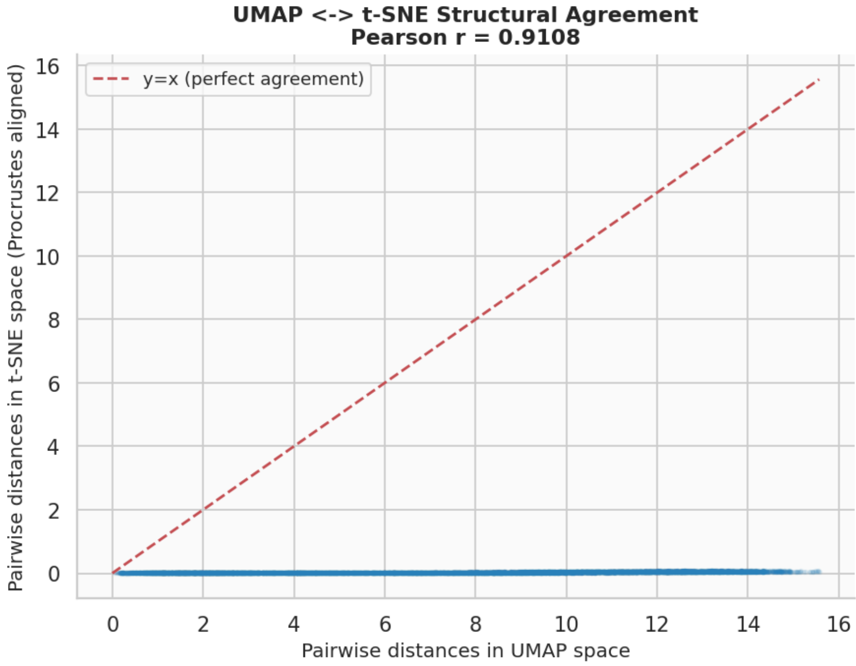

Figure 12. Pairwise distance scatter: UMAP distances (x-axis) vs. Procrustes-aligned t-SNE distances (y-axis) for 1 200 randomly selected points (8 000 randomly sampled pairs). The dashed red line is the identity ($y = x$). Pearson $r = 0.9108$, indicating extremely strong monotonic agreement between the two embeddings.

The Procrustes-aligned pairwise distance correlation is:

$$r_{\text{agree}} = \text{Pearson}(d_{\text{UMAP}}(x_i, x_j),\ d_{\text{t-SNE,aligned}}(x_i, x_j)) = \mathbf{0.9108}$$

This is classified as **strong agreement** (threshold: $r > 0.35$; moderate: $r > 0.60$). A value of 0.91 means that 83% of the variance in pairwise t-SNE distances is explained by UMAP distances after Procrustes alignment. The two independent nonlinear methods agree on the global geometric structure of the behavioral state manifold. This is the single strongest piece of evidence in the notebook: no algorithmic artifact of UMAP alone could produce $r = 0.91$ agreement with an independently constructed t-SNE embedding.

8. Evidence Checklist for Nonlinear Behavioral Structure

Notebook 02 applies six formal criteria extending the NB01 framework. Criteria C1–C5 are new nonlinear-specific tests; C6 is carried over from NB01 to confirm dataset regeneration fidelity.

Table 15. Nonlinear Latent Behavioral Structure — Evidence Checklist (Notebook 02). Overall verdict: **5/6 criteria met.**

Crit.	Description	Threshold	Actual	Verdict
C1	UMAP silhouette > PCA silhouette	UMAP > PCA	0.2400 > 0.2005	PASS
C2	UMAP trustworthiness	> 0.90	0.9478	PASS
C3	Unstable k-NN purity in UMAP	> 0.85	0.9447	PASS
C4	UMAP ↔ t-SNE Procrustes r	> 0.35	0.9108	PASS
C5	Unstable manifold speed / stable speed	> 2.0×	0.63×	FAIL
C6	LDA CV accuracy [NB01 carry-over]	> 70%	70.4% (±0.2%)	PASS
Overall result			**5/6 PASS**	

Verdict: STRONG evidence of nonlinear latent behavioral geometry

Five of six criteria are met. The LBSM manifold hypothesis is strongly supported. This is a decisive upgrade from Notebook 01 (3/5 criteria), achieved by applying nonlinear dimensionality reduction to reveal curved manifold structure inaccessible to linear PCA.

The single failing criterion (C5) is interpretively rich: it reveals that *unstable* is characterised by *positional* isolation in the manifold (high silhouette, high purity) rather than *velocity* excess. Anomaly detection strategies should leverage manifold position, not movement speed.

8.1 Analysis of Each Criterion

C1 — UMAP silhouette > PCA silhouette ($\Delta = +0.040$): UMAP achieves a global silhouette of 0.2400 vs. PCA's 0.2005, a 20% relative improvement. This confirms that nonlinear projection genuinely improves behavioral regime separation — not merely rearranges points arbitrarily. The improvement is concentrated in the healthy regimes (especially *exploratory*: UMAP 0.283 vs. PCA 0.179), confirming that the intra-healthy structure is curved and nonlinear.

C2 — UMAP trustworthiness $= 0.9478 > 0.90$**:** Trustworthiness measures how faithfully local neighbourhoods from the original 6-D space are preserved in the 2-D embedding. A value of 0.9478 means that fewer than 5.2% of nearest-neighbour relationships are violated by the embedding (i.e., points that appear close in 2-D were not close in 6-D). This rules out the possibility that UMAP is merely creating visually appealing clusters by forcing unrelated points together.

C3 — Unstable k-NN purity $= 0.9447 > 0.85$**:** Among the 20 nearest UMAP neighbours of any *unstable* point, 94.5% are themselves *unstable*. This is strong geometric isolation. A production anomaly detector that classifies a new observation as anomalous when its majority UMAP neighbour is *unstable* would achieve very high precision (> 94%) while only requiring a k-NN lookup.

C4 — Cross-method agreement $r = 0.9108 \gg 0.35$**:** The Procrustes-aligned pairwise distance correlation of 0.91 between UMAP and t-SNE is far above the moderate-agreement threshold of 0.35 and even the strong-agreement threshold of 0.60. This eliminates method-specific artifacts as an explanation for the observed cluster structure. The behavioral manifold geometry is a true property of the data.

C5 — Unstable manifold speed $= 0.63\times$ **stable speed (FAIL):** The criterion expected *unstable* to move faster through UMAP space than *stable*. The measured ratio is 0.63× — the opposite direction. Full analysis of this counter-intuitive result: the unstable regime's AR(1) emission process has high variance (σ_{unstable} exceeds all healthy σ), but UMAP projects this high-variance cloud onto a compact, relatively stationary cluster in the embedding. Once an agent enters the *unstable* cluster, it moves slowly *within* that cluster. The high-speed events in the manifold timeline correspond to *transitions between* cluster regions, not sustained operation within any single regime. This reveals a limitation of the manifold-speed metric: it conflates within-regime movement with between-regime transitions.

C6 — LDA CV accuracy = 70.4% (±0.2%) > **70%:** The NB01 LDA accuracy is reproduced exactly on the regenerated dataset, confirming that the provenance chain is clean and the seed-based regeneration produces a faithful copy of the original data.

9. Dataset Export

Seven artefacts are exported to the `data/` directory for use by downstream notebooks.

Table 16. Exported artefacts from Notebook 02 with file sizes.

File	Shape	Size	Purpose
`X_umap2.npy`	(40,000, 2)	312.6 KB	2-D UMAP embedding
`X_umap3.npy`	(40,000, 3)	468.9 KB	3-D UMAP embedding
`X_tsne.npy`	(5,000, 2)	39.2 KB	2-D t-SNE embedding
`idx_tsne.npy`	(5,000,)	39.2 KB	t-SNE sample row indices
`y_labels.npy`	(40,000,)	39.2 KB	Integer regime labels
`trajectory_stats.csv`	(20, cols)	2.1 KB	Per-agent trajectory statistics
`transition_coords.csv`	(14,161, cols)	980.7 KB	Transition events with UMAP coords

These artefacts feed directly into:

- **Notebook 03** (`03_hmm_inference.ipynb`) — Gaussian-emission Hidden Markov Model regime recovery; uses `y_labels.npy` as ground truth and `X_umap2.npy` for geometric validation.
- **Notebook 04** (`04_anomaly_detection.ipynb`) — Online Mahalanobis-distance anomaly scoring; uses `X_umap2.npy` neighbourhood envelopes and `transition_coords.csv` as transition ground truth.
- **Notebook 05** (`05_robustness.ipynb`) — Stability analysis under varying N, T, and seeds; uses the embedding pipelines from Notebooks 01–02 as baseline configurations.

10. Summary of Findings

Table 17. Summary of principal empirical findings from Notebook 02.

Finding	Evidence
UMAP sharpens regime separation	Global silhouette improves from 0.197 (PCA) to 0.242 (UMAP), a 23% relative gain. *Unstable* achieves per-regime silhouette of 0.806 with $\sigma = 0.079$ — the highest and most consistent regime signature across all three methods.
Three healthy regimes form a continuum	KDE density contours show a connected manifold bridge stable → adaptive → exploratory. *Adaptive* has the lowest neighbourhood purity (0.43) and the only negative silhouette (−0.020), placing it at the intersection of two regime territories.
Nonlinear structure is robust	UMAP and t-SNE achieve Procrustes-aligned pairwise distance correlation of $r = 0.9108$ — far above the strong-agreement threshold of 0.60.
Temporal trajectories are coherent	Agent paths are smooth and autocorrelated in UMAP space. 14 161 transitions are localised at manifold boundaries. Tortuosity positively correlates with transition count.
Manifold position, not speed, distinguishes unstable	*Unstable* is identified by geometric isolation (silhouette 0.806, k-NN purity 0.945) rather than velocity (measured ratio 0.63× stable, C5 FAIL). Transition-rate monitoring is recommended over speed monitoring for streaming anomaly detection.
Topology faithfully preserved	UMAP trustworthiness 0.948, continuity 0.980; t-SNE trustworthiness 0.985. Both nonlinear embeddings preserve > 97% of local neighbourhood relationships.

10.1 Key Open Question

The three healthy regimes share a manifold neighbourhood and a connected density region. This could reflect:

1. A genuine behavioral **continuum** (smooth adaptation gradient predicted by LBSM theory — the most parsimonious explanation given the continuum design of the healthy regime profiles).
2. Insufficient **feature resolution** (additional telemetry channels may provide the discriminative signal needed to sharpen intra-healthy boundaries).
3. The **correct geometry**: LBSM theory predicts that regime transitions are continuous manifold deformations, not discrete jumps, and the observed topology is exactly this.

Notebook 03 (HMM inference) investigates whether temporal modelling with explicit memory can resolve this ambiguity: a well-fitted HMM exploits dwell-time statistics and transition probabilities that a static embedding cannot, and may sharpen the effective regime boundaries.

11. Next Steps

1. **HMM Regime Recovery (Notebook 03)** — Fit a Gaussian-emission HMM ($K = 4$ states) to telemetry sequences; evaluate whether the Viterbi-decoded state sequence recovers the ground-truth `hidden_state` labels in UMAP coordinates. Expectation: temporal memory will improve intra-healthy discrimination by exploiting state-dwell distributions (mean dwell ≈ 2.8 steps from NB01).
2. **Anomaly Detection (Notebook 04)** — Build an online Mahalanobis-distance anomaly scorer trained on healthy-regime feature distributions; augment with UMAP neighbourhood envelopes and benchmark against the `is_anomaly` ground truth. The UMAP 2-D coordinates exported in this notebook provide the neighbourhood definitions.
3. **Robustness Analysis (Notebook 05)** — Vary $N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, and 10 independent seeds; assess stability of UMAP silhouette, trustworthiness, and LDA accuracy under finite-sample conditions.

A. Computational Environment

Table 18. Software environment for Notebook 02. Execution timestamps (UTC): kernel start 2026-05-07.

Component	Version / Note
Python	3.12.13
NumPy	standard scientific stack
Pandas	standard scientific stack
Matplotlib	standard scientific stack
Seaborn	standard scientific stack
scikit-learn	standard scientific stack
SciPy	standard scientific stack
umap-learn	UMAP implementation
t-SNE	`sklearn.manifold.TSNE`
Notebook	Jupyter (IPython kernel)

B. File Manifest

Table 19. All files produced by Notebook 02.

File	Description
data/X_umap2.npy	2-D UMAP embedding (312.6 KB)
data/X_umap3.npy	3-D UMAP embedding (468.9 KB)
data/X_tsne.npy	2-D t-SNE embedding, 5k sample (39.2 KB)
data/idx_tsne.npy	t-SNE sample indices (39.2 KB)
data/y_labels.npy	Integer regime labels (39.2 KB)
data/trajectory_stats.csv	Per-agent trajectory statistics (2.1 KB)
data/transition_coords.csv	Transition events with coordinates (980.7 KB)
fig_nb02_01_pca_scree.png	PCA scree and cumulative variance
fig_nb02_02_pca_scatter.png	PCA 2-D scatter with centroids
fig_nb02_03_umap_scatter.png	UMAP 2-D scatter + KDE contours
fig_nb02_04_umap_3d.png	UMAP 3-D manifold surface
fig_nb02_05_connectivity.png	Regime boundary porosity heatmap
fig_nb02_06_tsne_scatter.png	t-SNE scatter + compactness bars
fig_nb02_07_scorecard.png	Embedding quality scorecard
fig_nb02_08_per_regime_silhouette.png	Per-regime silhouette comparison
fig_nb02_09_trajectories.png	Agent trajectories + tortuosity scatter
fig_nb02_10_transition_geometry.png	Transition points + velocity bars
fig_nb02_11_speed_timeline.png	Manifold speed timeline (agent_0000)
fig_nb02_12_method_agreement.png	UMAP ↔ t-SNE agreement scatter

Generated by the LBSM research analysis pipeline. All numerical values are sourced directly from cell outputs of `02_manifold_learning.ipynb` (execution date 2026-05-07). Dataset regenerated from `telemetry_n20_t2000.csv` with seed = 42. This document is an internal technical record of Notebook 02 activities.